

Solution Architecture

Date	29-09-2026
Team ID	SWTID-2026-1297
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

Thunder Client



API Gateway / Load Balancer

Express.js Server



Search + Gemini AI Service

Auth Service *JWT*

Authentication + bcrypt

Core Logic Service

FAQ Management + FAQ



Data / Storage Layer

External API

Integrations

Google Gemini API



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation / Client Layer</i>	<i>Used to send API requests and test the backend endpoints.</i>	<i>Postman</i>
<i>API Gateway / Server</i>	<i>Receives HTTP requests and routes them to the appropriate API endpoints.</i>	<i>Node.js, Express.js</i>
<i>Authentication Service</i>	<i>Handles user authentication and protects authorized API routes.</i>	<i>JWT, bcrypt</i>
<i>Core Logic Service</i>	<i>Handles FAQ management, FAQ search and AI-powered answer generation.</i>	<i>Node.js, Express.js, Google Gemini AI</i>
<i>Database</i>	<i>Stores user and FAQ data.</i>	<i>MongoDB, Mongoose</i>
<i>External API</i>	<i>Provides AI-powered response generation.</i>	<i>Google Gemini API</i>